

PERSISTENCE OF THE TRILINEAR OPERATOR SPACE GROTHENDIECK OBSTRUCTION UNDER COMPLETE RETRACTIONS

DRAFT MANUSCRIPT

ABSTRACT. Briet and Palazuelos proved that the trilinear operator space Grothendieck inequality fails already on finite dimensional commutative C^* -algebras. We record a permanence principle showing that this obstruction passes through completely bounded retractions. Consequently the quantitative failure persists, with the same polynomial order up to constants, on full matrix algebras, on $B(H)$ for infinite dimensional H , and on every finite von Neumann algebra containing arbitrarily large finite dimensional abelian subalgebras with contractive conditional expectations. We also prove the corresponding lower bound for weighted symmetrized Haagerup-type envelopes: any uniformly bounded weighted modification must still fail on matrix algebras; a uniform weighted theorem would force the smallest weight on the commutative diagonal corner to decay polynomially. No conjectural statements are used.

1. INTRODUCTION

The bilinear operator space Grothendieck theorem says that the jointly completely bounded norm and the symmetrized completely bounded norm of a bilinear form on C^* -algebras are equivalent with universal constant at most 2; this is the theorem of Pisier-Shlyakhtenko and Haagerup-Musat. Pisier asked whether the analogous trilinear statement is true. Briet and Palazuelos answered negatively: there are trilinear forms on $(\ell_\infty^n)^3$ for which the ratio of the symmetrized completely bounded norm to the jointly completely bounded norm grows at least like a positive power of n .

The negative result is sometimes read as a purely commutative obstruction. This note shows that the obstruction is not confined to the commutative category. The simple mechanism is complete retraction. If three operator spaces contain completely complemented copies of the spaces on which the trilinear obstruction occurs, then the obstruction extends to the larger triple. Since the diagonal algebra $D_n \cong \ell_\infty^n$ is the range of a completely positive contractive projection from M_n , the Briet-Palazuelos forms extend from D_n to M_n without losing their asymptotic separation. Thus full matrix algebras, although finite dimensional, nuclear, exact, and simple, cannot support an unrestricted trilinear operator space Grothendieck theorem.

The second point concerns modified symmetrized norms. A natural response to the failure theorem is to replace the six-fold symmetrized norm by a weighted version. We prove a necessary calibration condition for any such weighted theory. If the smallest weight stays bounded below on diagonal corners, then the same polynomial obstruction survives on matrix algebras. Hence a weighted theorem uniform over matrix algebras must make at least one ordering polynomially cheap on the commutative diagonal.

2. TRILINEAR CONSTANTS

Let E_1, E_2, E_3 be operator spaces and let

$$u : E_1 \times E_2 \times E_3 \longrightarrow \mathbb{C}$$

be a trilinear form. The jointly completely bounded amplification is

$$u_d^{\text{jcb}} \left(\sum_i x_i^1 \otimes a_i^1, \sum_j x_j^2 \otimes a_j^2, \sum_k x_k^3 \otimes a_k^3 \right) = \sum_{i,j,k} u(x_i^1, x_j^2, x_k^3) a_i^1 \otimes a_j^2 \otimes a_k^3$$

as an element of $M_d \otimes M_d \otimes M_d$. Put

$$\|u\|_{\text{jcb}} = \sup_d \|u_d^{\text{jcb}}\|.$$

For an ordered trilinear form $v : E_1 \times E_2 \times E_3 \rightarrow \mathbb{C}$ define

$$\|v\|_{\text{cb}} = \sup_d \left\| \sum_{i,j,k} v(x_i^1, x_j^2, x_k^3) a_i^1 a_j^2 a_k^3 \right\|,$$

where the supremum is over contractions in $M_d(E_1)$, $M_d(E_2)$, and $M_d(E_3)$. For $\pi \in S_3$ write

$$u^\pi(x_{\pi(1)}, x_{\pi(2)}, x_{\pi(3)}) = u(x_1, x_2, x_3).$$

Definition 2.1. The symmetrized completely bounded norm of u is

$$\|u\|_{\text{sym}} = \inf \left\{ \sum_{\pi \in S_3} \|v_\pi^\pi\|_{\text{cb}} : u = \sum_{\pi \in S_3} v_\pi \right\}.$$

The trilinear operator space Grothendieck constant of the triple (E_1, E_2, E_3) is

$$\text{GT}(E_1, E_2, E_3) = \sup_{u \neq 0} \frac{\|u\|_{\text{sym}}}{\|u\|_{\text{jcb}}}.$$

The inequality $\|u\|_{\text{jcb}} \leq \|u\|_{\text{sym}}$ follows from the standard commuting representation trick: in the ordered amplification use $a_i^1 \otimes I \otimes I$, $I \otimes a_j^2 \otimes I$, and $I \otimes I \otimes a_k^3$.

Definition 2.2. Let $\omega = (\omega_\pi)_{\pi \in S_3}$ be positive weights. The weighted symmetrized norm is

$$\|u\|_{\text{sym}, \omega} = \inf \left\{ \sum_{\pi \in S_3} \omega_\pi \|v_\pi^\pi\|_{\text{cb}} : u = \sum_{\pi \in S_3} v_\pi \right\}.$$

Set

$$\text{GT}_\omega(E_1, E_2, E_3) = \sup_{u \neq 0} \frac{\|u\|_{\text{sym}, \omega}}{\|u\|_{\text{jcb}}}, \quad \mathfrak{m}(\omega) = \min_{\pi \in S_3} \omega_\pi.$$

Since

$$\|u\|_{\text{sym}, \omega} \geq \mathfrak{m}(\omega) \|u\|_{\text{sym}},$$

all lower bounds for GT immediately give weighted lower bounds after multiplication by $\mathfrak{m}(\omega)$.

3. COMPLETE RETRACTIONS

Definition 3.1. Let F and E be operator spaces. We say that F is completely λ -retracted in E if there are completely bounded maps

$$j : F \rightarrow E, \quad P : E \rightarrow F$$

such that $Pj = \text{id}_F$ and

$$\|j\|_{\text{cb}} \|P\|_{\text{cb}} \leq \lambda.$$

If $\lambda = 1$, we say completely contractively retracted.

Theorem 3.2 (Retraction monotonicity). *Let F_i be completely λ_i -retracted in E_i for $i = 1, 2, 3$. Then*

$$\text{GT}(E_1, E_2, E_3) \geq \frac{\text{GT}(F_1, F_2, F_3)}{\lambda_1 \lambda_2 \lambda_3}.$$

More generally, for every positive weight vector ω ,

$$\text{GT}_\omega(E_1, E_2, E_3) \geq \frac{\text{GT}_\omega(F_1, F_2, F_3)}{\lambda_1 \lambda_2 \lambda_3}.$$

Proof. Choose maps $j_i : F_i \rightarrow E_i$ and $P_i : E_i \rightarrow F_i$ with $P_i j_i = \text{id}_{F_i}$. Let $u : F_1 \times F_2 \times F_3 \rightarrow \mathbb{C}$ be a trilinear form and define its extension

$$\tilde{u}(x_1, x_2, x_3) = u(P_1 x_1, P_2 x_2, P_3 x_3).$$

Functoriality of the jointly completely bounded amplification gives

$$\|\tilde{u}\|_{\text{jcb}} \leq \|P_1\|_{\text{cb}} \|P_2\|_{\text{cb}} \|P_3\|_{\text{cb}} \|u\|_{\text{jcb}}.$$

Since

$$u(y_1, y_2, y_3) = \tilde{u}(j_1 y_1, j_2 y_2, j_3 y_3),$$

the same functoriality for each ordered completely bounded norm gives

$$\|u\|_{\text{sym}, \omega} \leq \|j_1\|_{\text{cb}} \|j_2\|_{\text{cb}} \|j_3\|_{\text{cb}} \|\tilde{u}\|_{\text{sym}, \omega}.$$

Therefore

$$\frac{\|\tilde{u}\|_{\text{sym}, \omega}}{\|\tilde{u}\|_{\text{jcb}}} \geq \frac{1}{\prod_i \|j_i\|_{\text{cb}} \|P_i\|_{\text{cb}}} \frac{\|u\|_{\text{sym}, \omega}}{\|u\|_{\text{jcb}}}.$$

Taking the supremum over u and then the infimum over the retraction constants proves the weighted statement. The unweighted statement is the case $\omega_\pi = 1$. $\square$

4. MATRIX ALGEBRAS

Let $D_n \subset M_n$ be the diagonal algebra. The inclusion $D_n \hookrightarrow M_n$ is a complete isometry and the diagonal projection

$$\mathbb{E}_n((a_{ij})_{i,j=1}^n) = \text{diag}(a_{11}, \dots, a_{nn})$$

is a completely positive contraction. Hence D_n is completely contractively retracted in M_n .

We use the following quantitative theorem of Briet and Palazuelos.

Theorem 4.1 (Briet-Palazuelos). *There are constants $a > 0$ and $\eta > 0$ such that for every $n \geq 2$ there is a trilinear form $u_n : (\ell_\infty^n)^3 \rightarrow \mathbb{C}$ satisfying*

$$\|u_n\|_{\text{sym}} \geq a n^\eta \|u_n\|_{\text{jcb}}.$$

In their construction one may take $\eta = 1/8$ up to absolute changes in the constant.

Theorem 4.2 (Failure on full matrix algebras). *There are constants $a > 0$ and $\eta > 0$ such that, for every $n \geq 2$,*

$$\text{GT}(M_n, M_n, M_n) \geq a n^\eta.$$

Moreover, for every positive weight vector ω ,

$$\text{GT}_\omega(M_n, M_n, M_n) \geq a \mathfrak{m}(\omega) n^\eta.$$

Proof. Identify ℓ_∞^n with D_n . By Theorem 4.1,

$$\text{GT}(D_n, D_n, D_n) \geq an^\eta.$$

Since D_n is completely contractively retracted in M_n , Theorem 3.2 gives the unweighted estimate. For the weighted estimate, use

$$\text{GT}_\omega(D_n, D_n, D_n) \geq m(\omega)\text{GT}(D_n, D_n, D_n)$$

and apply Theorem 3.2 again. $\square$

Corollary 4.3. *No trilinear operator space Grothendieck inequality with a constant depending only on exactness, nuclearity, or finite dimensionality can hold for C^* -algebras. In particular, the obstruction persists on the simple nuclear exact algebras M_n with polynomial growth in n .*

Proof. Each M_n is finite dimensional, nuclear, exact, and simple. A bound depending only on any of these qualitative properties would give a uniform upper bound for $\text{GT}(M_n, M_n, M_n)$, contradicting Theorem 4.2. $\square$

Corollary 4.4 (Calibration of weighted modifications). *Let $\omega^{(n)}$ be positive weights assigned to triples (M_n, M_n, M_n) . If*

$$\sup_{n \geq 2} \text{GT}_{\omega^{(n)}}(M_n, M_n, M_n) < \infty,$$

then

$$m(\omega^{(n)}) = O(n^{-\eta})$$

with η as in Theorem 4.1. In particular, every weighted theory whose weights are uniformly bounded below fails uniformly on matrix algebras.

Proof. This follows immediately from the weighted part of Theorem 4.2. $\square$

5. INFINITE DIMENSIONAL CONSEQUENCES

The finite dimensional estimates imply unbounded trilinear constants for many standard infinite dimensional operator algebras.

Corollary 5.1 (The algebra $B(H)$). *If H is infinite dimensional, then*

$$\text{GT}(B(H), B(H), B(H)) = \infty.$$

The same conclusion holds for GT_ω whenever the weights are bounded below by a positive constant.

Proof. For every n , choose orthonormal vectors $e_1, \dots, e_n$ in H and let D_n be the diagonal algebra supported on their span. The map

$$T \mapsto \sum_{k=1}^n \langle Te_k, e_k \rangle e_k \otimes e_k^*$$

is completely positive and contractive onto D_n , and the inclusion of D_n into $B(H)$ is completely isometric. Thus D_n is completely contractively retracted in $B(H)$ for every n . Apply Theorems 3.2 and 4.1 and let $n \rightarrow \infty$. $\square$

Corollary 5.2 (Finite von Neumann algebras). *Let M be a finite von Neumann algebra with a faithful normal trace. Suppose that for arbitrarily large n the algebra M contains a unital abelian subalgebra generated by n nonzero orthogonal projections. Then*

$$\text{GT}(M, M, M) = \infty.$$

In particular this holds for diffuse finite von Neumann algebras and for type II_1 factors.

Proof. In a finite von Neumann algebra, the trace-preserving conditional expectation onto a von Neumann subalgebra is completely positive and contractive. Hence each finite dimensional abelian subalgebra $D_n \cong \ell_\infty^n$ appearing in the hypothesis is completely contractively retracted in M . The conclusion follows from Theorems 3.2 and 4.1. $\square$

6. CONCLUSION

The trilinear obstruction is stable under completely bounded retractions. Therefore exactness, nuclearity, noncommutativity, and passage from diagonal algebras to full matrix algebras do not repair the failed trilinear operator space Grothendieck theorem. Any viable replacement must exclude large completely complemented commutative corners or weaken the symmetrized Haagerup envelope by weights that degenerate at least polynomially on such corners.

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